Collections

1.LIST

1. Create a list of 10 student marks and find the highest mark.
2. Find the lowest and highest salary from an employee salary list.
3. Find the sum and average of numbers in a list.
4. Separate a list of numbers into even and odd numbers.
5. Remove duplicate values from a list.
6. Find the second largest number without using sort().
7. Reverse a list without using reverse().
8. Find common elements between two lists.
9. Create a shopping cart list and allow the user to add, remove, and view products.
10. Find how many times a particular item occurs in a list.

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2. TUPLE

11. Create a tuple of 5 student names and print each name using a loop.
12. Create a tuple of numbers and find the maximum, minimum, and sum.
13. Count how many times a particular value occurs in a tuple.
14. Find the index position of a given value in a tuple.
15. Convert a tuple into a list, add new values, and convert it back to a tuple.
16. Create a tuple containing student details (name, age, mark) and unpack the values.

17. Given a tuple of marks, count students who scored above 80.

18. Create two tuples and concatenate them into a single tuple.

19. Check whether a particular item exists in a tuple.

20. Given a tuple of numbers, create a new tuple containing only even numbers.

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3. SET

21. Create a set from a list containing duplicate values and remove duplicates.

22. Find the common students between two sets.

23. Find students who are present in Set A but not Set B.

24. Find all unique students from two class sets.

25. Check whether one set is a subset of another set.

26. Check whether two sets are completely different using disjoint sets.

27. Create a set of numbers from 1–20 and remove all even numbers.

28. Find the union, intersection, and difference of two sets.

29. Given two sets of employees from different departments, find employees working in both departments.

30. Create a set of purchased product IDs and check whether a given product ID was purchased.

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4. DICTIONARY

31. Create a dictionary of student names and marks and find the student with the highest mark.

32. Create an employee dictionary containing name, salary, and department and display employees earning above ₹30,000.

33. Count the frequency of each character in a string using a dictionary.

34. Count the frequency of each word in a sentence.

35. Create a product-price dictionary and allow the user to search for a product price.

36. Create a dictionary containing numbers from 1–10 and their squares.

37. Merge two dictionaries into one dictionary.

38. Create a student dictionary and separate students into **Pass** and **Fail** based on marks.

39. Create a phone book using a dictionary with **Add, Search, Update, and Delete** operations.

40. Create a shopping cart using a dictionary where the **product is the key and quantity is the value**, then calculate the total quantity.